Gradient Networks for Universal Magnetic Modeling of Synchronous Machines

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arXiv:2602.14947v4 [eess.SY] 24 Sep 2026

***Abstract*—This paper presents a physics-constrained neural network framework for magnetic modeling of saturable synchronous machines, including spatial harmonics. By embedding gradient networks into the machine equations to model conservative electromagnetic behavior, the framework satisfies reciprocity and energy conservation by construction, while universally approximating any physically feasible magnetic characteristic. Unlike lookup tables and black-box neural networks, it guarantees monotonicity, invertibility, and smooth outputs, and remains highly data efficient. The method is validated using measured and finite-element method (FEM) data from a 5.6-kW permanent-magnet (PM) synchronous reluctance machine, and is demonstrated in real-time closed-loop control on an embedded platform. The results confirm accurate, physically consistent, and computationally efficient performance.**

***Index Terms*—Dynamic modeling, electric drives, Hamiltonian, magnetic saturation, neural networks, physics-informed machine learning, synchronous machines.**

I. Introduction

DYNAMIC models of synchronous machines are essential for control, estimation, monitoring, and optimization. A particularly challenging aspect is the magnetic model, which describes the relationship between flux linkages, currents, rotor angle, and electromagnetic torque [1], [2]. In modern power-dense designs, such as permanent-magnet (PM) and synchronous reluctance machines, magnetic saturation and cross-saturation effects are significant and must often be incorporated into control models. Spatial harmonics, on the other hand, can be included in high-fidelity models for time-domain simulations during the design stage.

The nonlinear magnetic behavior is typically represented by analytical functions [3]–[7], semi-analytical models [8], lookup tables [9]–[13], or neural networks [14]–[18]. These models can be characterized from finite-element method (FEM) data [19], laboratory measurements [13], [20], or automatic identification routines [21]. Analytical models are limited by their functional form and are difficult to extend to higher dimensions (e.g., for spatial harmonics or multi-phase machines). Lookup tables suffer from the curse of dimensionality, high memory requirements, and non-smooth outputs under linear interpolation. Black-box neural networks [15]–[17] can approximate high-dimensional complex maps and require less memory than lookup tables, but they require large training datasets and do not guarantee reciprocity [22] or invertibility. Although some architectures incorporate spatial periodicity [14], [18], neither reciprocity nor monotonicity is guaranteed by construction.

Physics-informed neural networks address these limitations by embedding known physical principles [23]–[26]. Electric machines are port-Hamiltonian systems [27], in which the current and electromagnetic torque are gradients of the magnetic field energy with respect to the flux linkage and rotor angle, respectively [1], [2]. Hamiltonian neural networks [25], [26] could exploit this structure by modeling the energy as a scalar network and differentiating it, which improves data efficiency and physical consistency. However, they become harder to optimize as the input dimension grows, lack a universal approximation guarantee for the gradient field [28], and require second-order backpropagation in training and network differentiation at inference. A similar differentiation principle was applied to electromagnetic torque modeling [18], but the relationship between currents and flux linkages was not considered.

In this paper, we propose a physics-constrained magnetic modeling framework that embeds gradient networks [28] into the fundamental electromechanical equations of synchronous machines [1], [2]. Gradient networks directly model monotone gradient fields without differentiating a scalar network, avoiding these training and inference costs. The main contributions are:

- an energy-based magnetic modeling framework in which current and torque are derived from the gradient of a field energy function, guaranteeing reciprocity and monotonicity (and thus invertibility) by construction;
- a co-energy-based formulation for directly modeling flux linkage and torque with the same physical guarantees;
- architectural encodings of physical structure, enforcing q-axis symmetry via energy symmetrization and capturing spatial harmonics via Fourier features [29];
- a computationally efficient p-norm gradient activation, together with guidance for selecting activation functions for current and flux-linkage maps;
- experimental validation on two PM synchronous reluctance machines (5.6 and 28 kW) alongside FEM validation, demonstrating data efficiency and real-time deployment in a closed-loop embedded drive system.

This paper is organized as follows. Section II reviews the

This work was supported in part by the ABB Oy, in part by the Aalto University House of AI, and in part by the Research Council of Finland Centre of Excellence in High-Speed Electromechanical Energy Conversion Systems. The authors acknowledge the use of EPE infrastructure of Aalto School of Electrical Engineering.

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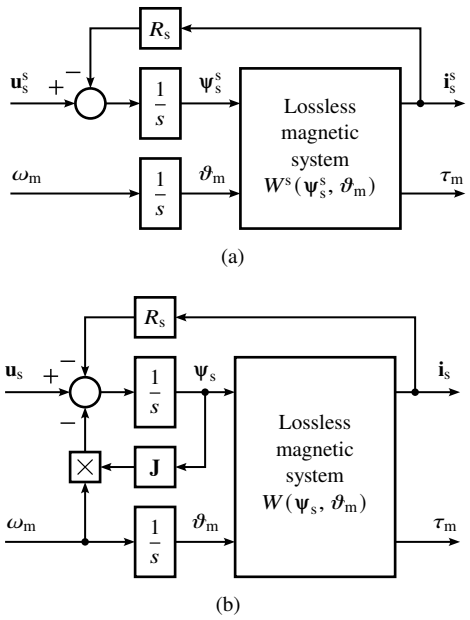


Fig. 1. Electromechanical dynamics of a generic synchronous machine: (a) stator coordinates; (b) rotor coordinates. The blocks $1/s$ denote integration in time.

machine model. Section III details the proposed architecture. Section IV presents offline validation and real-time control results. Section V concludes the paper.

II. Physics-Based Machine Model

This section describes the fundamental dynamic model of a generic synchronous machine [1], [2]. Scalars are denoted by italic letters, column vectors by bold lowercase letters, and matrices by bold uppercase letters. All quantities are in per-unit unless otherwise specified.

A. Stator Coordinates

The synchronous machine model in stator ($\alpha\beta$) coordinates is first considered. The state equations are

$$\frac{\mathrm{d}\boldsymbol{\psi}_s^s}{\mathrm{d}t} = \mathbf{u}_s^s - R_s\mathbf{i}_s^s \tag{1a}$$

$$\frac{\mathrm{d}\vartheta_m}{\mathrm{d}t} = \omega_m \tag{1b}$$

where $\boldsymbol{\psi}_s^s = [\psi_\alpha, \psi_\beta]^\top$ is the stator flux-linkage vector, $\mathbf{i}_s^s = [i_\alpha, i_\beta]^\top$ is the stator current vector, $\mathbf{u}_s^s = [u_\alpha, u_\beta]^\top$ is the stator voltage vector, and R_s is the stator resistance. Furthermore, ϑ_m is the electrical angle of the rotor d-axis with respect to the stator coordinates, and ω_m is the electrical angular speed of the rotor.

Assuming a lossless (conservative) magnetic field system, the magnetic behavior is fully described by its field energy function $W^s(\boldsymbol{\psi}_s^s, \vartheta_m)$. Hence, the stator current and electromagnetic torque are given by [1], [2]

$$\mathbf{i}_s^s = \left[\frac{\partial W^s(\boldsymbol{\psi}_s^s, \vartheta_m)}{\partial \boldsymbol{\psi}_s^s}\right]^\top \tag{2a}$$

$$\tau_m = -\frac{\partial W^s(\boldsymbol{\psi}_s^s, \vartheta_m)}{\partial \vartheta_m}. \tag{2b}$$

The minus sign appears in the torque expression since the positive mechanical power is defined out of the machine. It can be seen that the current and electromagnetic torque (with negative sign) constitute the gradient of the field energy function [27], [30]. Fig. 1(a) shows the block diagram corresponding to (1) and (2).

If needed, core losses can be included outside the magnetic model by adding an equivalent core-loss resistance (or a more general nonlinear dissipative element) in parallel to the main magnetic circuit, as is standard practice [8], [9]. This reduces the current entering the lossless part of the model, but otherwise the model remains unchanged.

B. Rotor Coordinates

The machine model is more convenient to express in rotor (dq) coordinates, which rotate with the rotor d-axis. Coordinate transformation to rotor coordinates can be expressed using the matrix exponential as

$$\boldsymbol{\psi}_s = \begin{bmatrix} \cos\vartheta_m & \sin\vartheta_m \\ -\sin\vartheta_m & \cos\vartheta_m \end{bmatrix} \boldsymbol{\psi}_s^s = \mathrm{e}^{-\vartheta_m\mathbf{J}}\boldsymbol{\psi}_s^s \tag{3}$$

where $\mathbf{J} = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ is the orthogonal rotation matrix. The flux-linkage vector $\boldsymbol{\psi}_s = [\psi_d, \psi_q]^\top$ is used as an example, but other vectors are transformed similarly.

1) Model Structure: Using (3), the state equations (1) are transformed to rotor coordinates as

$$\frac{\mathrm{d}\boldsymbol{\psi}_s}{\mathrm{d}t} = \mathbf{u}_s - R_s\mathbf{i}_s - \omega_m\mathbf{J}\boldsymbol{\psi}_s \tag{4a}$$

$$\frac{\mathrm{d}\vartheta_m}{\mathrm{d}t} = \omega_m \tag{4b}$$

where $\mathbf{i}_s = [i_d, i_q]^\top$ is the current vector and $\mathbf{u}_s = [u_d, u_q]^\top$ is the voltage vector. The field energy function can be expressed in rotor coordinates, $W(\boldsymbol{\psi}_s, \vartheta_m) = W^s(\boldsymbol{\psi}_s^s, \vartheta_m)$. Applying the coordinate transformation and the chain rule (see Appendix A), the magnetic model (2) becomes

$$\mathbf{i}_s = \left[\frac{\partial W(\boldsymbol{\psi}_s, \vartheta_m)}{\partial \boldsymbol{\psi}_s}\right]^\top \tag{5a}$$

$$\tau_m = \mathbf{i}_s^\top\mathbf{J}\boldsymbol{\psi}_s - \frac{\partial W(\boldsymbol{\psi}_s, \vartheta_m)}{\partial \vartheta_m}. \tag{5b}$$

The gradient relation (5a) defines the current map $\mathbf{i}_s(\boldsymbol{\psi}_s, \vartheta_m)$. Fig. 1(b) shows the corresponding block diagram.

In magnetic systems, increasing flux linkage requires increasing current [1]. Hence, the current map is strictly monotone, meaning that the incremental inverse inductance matrix $\boldsymbol{\Gamma}_s = \partial\mathbf{i}_s/\partial\boldsymbol{\psi}_s$ is positive definite. This corresponds to a strictly convex field energy $W(\boldsymbol{\psi}_s, \vartheta_m)$ with respect to $\boldsymbol{\psi}_s$, guaranteeing unique invertibility between flux linkages and currents.

If spatial harmonics are omitted, the field energy depends only on the flux linkage, $W(\boldsymbol{\psi}_s)$, simplifying the current map to $\mathbf{i}_s(\boldsymbol{\psi}_s)$ and the electromagnetic torque to $\tau_m = \mathbf{i}_s^\top\mathbf{J}\boldsymbol{\psi}_s$. As a familiar special case, this formulation includes the standard magnetically linear machine, where the field energy is quadratic, $W(\boldsymbol{\psi}_s) = \frac{1}{2}(\boldsymbol{\psi}_s - \boldsymbol{\psi}_f)^\top\boldsymbol{\Gamma}_s(\boldsymbol{\psi}_s - \boldsymbol{\psi}_f)$, with the constant inverse inductance matrix $\boldsymbol{\Gamma}_s = \mathrm{diag}(1/L_d, 1/L_q)$ and the PM-flux vector $\boldsymbol{\psi}_f = [\psi_f, 0]^\top$. In this case, (5a) reduces to the linear relationship $\mathbf{i}_s = \boldsymbol{\Gamma}_s(\boldsymbol{\psi}_s - \boldsymbol{\psi}_f)$.

2) Co-Energy-Based Model: Alternatively, a dual model can be derived via the Legendre transform $W(\boldsymbol{\psi}_\mathrm{s}, \vartheta_\mathrm{m}) + W'(\mathbf{i}_\mathrm{s}, \vartheta_\mathrm{m}) = \mathbf{i}_\mathrm{s}^\top \boldsymbol{\psi}_\mathrm{s}$, which relates the field energy W and co-energy W' [1], [2]. This yields

$$\boldsymbol{\psi}_\mathrm{s} = \left[\frac{\partial W'(\mathbf{i}_\mathrm{s}, \vartheta_\mathrm{m})}{\partial \mathbf{i}_\mathrm{s}}\right]^\top \tag{6a}$$

$$\tau_\mathrm{m} = \mathbf{i}_\mathrm{s}^\top \mathbf{J} \boldsymbol{\psi}_\mathrm{s} + \frac{\partial W'(\mathbf{i}_\mathrm{s}, \vartheta_\mathrm{m})}{\partial \vartheta_\mathrm{m}}. \tag{6b}$$

Note that the sign of the partial derivative term in the torque expression is positive. Similarly, the gradient relation in (6) defines the flux-linkage map $\boldsymbol{\psi}_\mathrm{s}(\mathbf{i}_\mathrm{s}, \vartheta_\mathrm{m})$, which simplifies to $\boldsymbol{\psi}_\mathrm{s}(\mathbf{i}_\mathrm{s})$ without spatial harmonics. The choice between the energy-based model (5) and the co-energy-based model (6) depends on whether flux linkage or current is chosen as the input in a given application.

3) Rotor-Angle Representation: Due to geometrical symmetries, spatial harmonics cause the magnetic characteristics to repeat k times per electrical revolution, with a fundamental period of $2\pi/k$. To enforce this periodicity and avoid artificial discontinuities at the period boundaries, the rotor angle is represented by the periodic unit vector

$$\boldsymbol{\vartheta} = \begin{bmatrix}\cos(k\vartheta_\mathrm{m}) \\ \sin(k\vartheta_\mathrm{m})\end{bmatrix}. \tag{7a}$$

In machine learning, such sinusoidal encodings are often called Fourier features [29]. Expressing the field energy as $W(\boldsymbol{\psi}_\mathrm{s}, \vartheta_\mathrm{m}) = \tilde{W}(\boldsymbol{\psi}_\mathrm{s}, \boldsymbol{\vartheta})$ and applying the chain rule, the model (5) becomes

$$\mathbf{i}_\mathrm{s} = \left[\frac{\partial \tilde{W}(\boldsymbol{\psi}_\mathrm{s}, \boldsymbol{\vartheta})}{\partial \boldsymbol{\psi}_\mathrm{s}}\right]^\top \tag{7b}$$

$$\tau_\mathrm{m} = \mathbf{i}_\mathrm{s}^\top \mathbf{J} \boldsymbol{\psi}_\mathrm{s} + k\boldsymbol{\vartheta}^\top \mathbf{J} \left[\frac{\partial \tilde{W}(\boldsymbol{\psi}_\mathrm{s}, \boldsymbol{\vartheta})}{\partial \boldsymbol{\vartheta}}\right]^\top. \tag{7c}$$

The analogous transformation applies to the co-energy-based model (6).

4) Symmetries From the Magnetic Geometry: Assuming the d-axis aligned with the PM flux, the magnetic geometry is symmetric about the d-axis. Consequently, reversing the signs of the q-axis flux linkage and rotor angle negates the q-axis current but leaves the d-axis current unchanged [13]

$$\mathbf{i}_\mathrm{s}(\mathbf{C}\boldsymbol{\psi}_\mathrm{s}, \mathbf{C}\boldsymbol{\vartheta}) = \mathbf{C}\mathbf{i}_\mathrm{s}(\boldsymbol{\psi}_\mathrm{s}, \boldsymbol{\vartheta}) \qquad \mathbf{C} = \mathrm{diag}(1, -1) \tag{8}$$

where $\mathbf{C}$ negates the second component of a vector. This implies that the field energy is an even function with respect to this reflection, $\tilde{W}(\boldsymbol{\psi}_\mathrm{s}, \boldsymbol{\vartheta}) = \tilde{W}(\mathbf{C}\boldsymbol{\psi}_\mathrm{s}, \mathbf{C}\boldsymbol{\vartheta})$. To enforce this symmetry by construction, the energy function is symmetrized as

$$\overline{W}(\boldsymbol{\psi}_\mathrm{s}, \boldsymbol{\vartheta}) = \frac{1}{2}\left[\tilde{W}(\boldsymbol{\psi}_\mathrm{s}, \boldsymbol{\vartheta}) + \tilde{W}(\mathbf{C}\boldsymbol{\psi}_\mathrm{s}, \mathbf{C}\boldsymbol{\vartheta})\right]. \tag{9}$$

Replacing $\tilde{W}$ with $\overline{W}$ in (7) guarantees the symmetry (8). The same formulation applies to the co-energy function $\tilde{W}'(\mathbf{i}_\mathrm{s}, \boldsymbol{\vartheta})$. Without spatial harmonics, the angle dependence is dropped, and (8) reduces to the well-known relations $i_\mathrm{d}(\psi_\mathrm{d}, -\psi_\mathrm{q}) = i_\mathrm{d}(\psi_\mathrm{d}, \psi_\mathrm{q})$ and $i_\mathrm{q}(\psi_\mathrm{d}, -\psi_\mathrm{q}) = -i_\mathrm{q}(\psi_\mathrm{d}, \psi_\mathrm{q})$ [20].

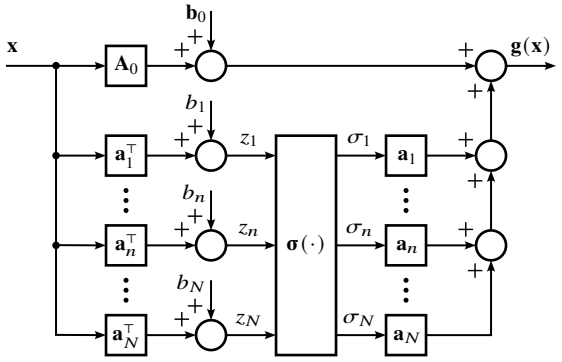


Fig. 2. Gradient network used for the magnetic models.

III. Proposed Gradient-Network Magnetic Models

A. Gradient Networks

1) Structure: Gradient networks can universally approximate any continuous gradient field [28]. Furthermore, they can be parameterized to guarantee monotonicity, which is the relevant case for the proposed magnetic models. We use an architecture with N hidden units, illustrated in Fig. 2, given by

$$\mathbf{z} = \mathbf{A}\mathbf{x} + \mathbf{b} \tag{10a}$$

$$\mathbf{g}(\mathbf{x}) = \mathbf{A}_0\mathbf{x} + \mathbf{b}_0 + \mathbf{A}^\top \boldsymbol{\sigma}(\mathbf{z}) \tag{10b}$$

where $\mathbf{A} = [\mathbf{a}_1 \dots \mathbf{a}_N]^\top$ is the weight matrix, $\mathbf{b} = [b_1 \dots b_N]^\top$ is the bias vector, $\boldsymbol{\sigma}(\mathbf{z})$ is the activation function, and $\mathbf{b}_0$ is the output bias.

The network (10) is conservative by construction: its Jacobian $\mathbf{J}_\mathbf{g} = \partial\mathbf{g}/\partial\mathbf{x}$ is symmetric whenever $\mathbf{A}_0$ and the activation Jacobian $\mathbf{J}_{\boldsymbol{\sigma}} = \partial\boldsymbol{\sigma}/\partial\mathbf{z}$ are symmetric (see Appendix B). To also guarantee monotonicity, $\mathbf{A}_0$ and $\mathbf{J}_{\boldsymbol{\sigma}}$ are chosen to be positive semidefinite. For elementwise activations $\boldsymbol{\sigma}(\mathbf{z}) = [\sigma_1(z_1) \dots \sigma_N(z_N)]^\top$, this simply requires each σ_n to be non-decreasing. Consequently, $\mathbf{g}(\mathbf{x})$ represents the gradient of an underlying convex energy function $W(\mathbf{x})$, i.e., $\mathbf{g}(\mathbf{x}) = [\partial W(\mathbf{x})/\partial\mathbf{x}]^\top$, without the need to model or differentiate $W(\mathbf{x})$ explicitly.

2) Elementwise Activations: Elementwise activations apply a scalar function $\sigma(z)$ independently to each hidden unit. While computationally efficient, they represent only a subset of monotone gradient fields and lack universal approximation guarantees [28].

To capture magnetic saturation, the activation type must match the curvature of the modeled relationship. In the current map $\mathbf{i}_\mathrm{s}(\boldsymbol{\psi}_\mathrm{s})$, saturation causes the current to rise progressively with increasing flux linkage [31]. This behavior is matched by rectifier-type activations with a non-decreasing derivative, such as the algebraic squareplus [32]

$$\sigma(z) = \frac{1}{2}\left(z + \sqrt{z^2 + \beta}\right) \tag{11}$$

where the positive parameter β tunes the smoothness around zero. As shown in Fig. 3, its derivative $\mathrm{d}\sigma/\mathrm{d}z = (1 + z/\sqrt{z^2+\beta})/2$ smoothly transitions from zero to one. Compared to the softplus, squareplus avoids exponential functions and is computationally more efficient.

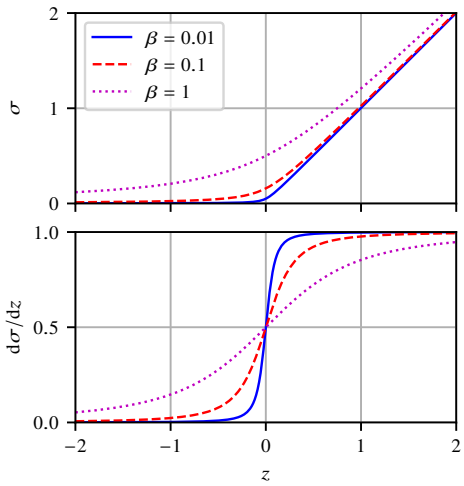


Fig. 3. Elementwise squareplus activation σ in (11) and its derivative $\mathrm{d}\sigma/\mathrm{d}x$ at different values of parameter β. The shape of the algebraic sigmoid (12) is the same as the derivative of the squareplus, but shifted vertically and scaled.

Conversely, in the flux-linkage map $\boldsymbol{\psi}_\mathrm{s}(\mathbf{i}_\mathrm{s})$, the flux linkage saturates with increasing current, matching the familiar flattening of magnetic saturation curves. This behavior suits sigmoid-type activations. Here, we use the algebraic sigmoid

$$\sigma(z) = \frac{z}{\sqrt{z^2+\beta}} \tag{12}$$

which corresponds to a vertically shifted and scaled version of $\mathrm{d}\sigma/\mathrm{d}z$ in (11).

3) Vector Activations: Unlike elementwise activations, vector activations produce outputs where each component depends on the entire input vector $\mathbf{z}$. A standard vector activation is the scaled softmax, given by

$$\boldsymbol{\sigma}(\mathbf{z}) = \frac{1}{\sum_{n=1}^{N} \mathrm{e}^{\beta z_n}} \begin{bmatrix} \mathrm{e}^{\beta z_1} \\ \vdots \\ \mathrm{e}^{\beta z_N} \end{bmatrix} \tag{13}$$

where the positive learnable parameter β scales the inputs. Fig. 4(a) illustrates the first component $\sigma_1(z_1, z_2)$ for the two-dimensional case. The softmax is the gradient of the convex log-sum-exp function $S(\mathbf{z}) = \log(\sum_{n=1}^{N} \mathrm{e}^{\beta z_n})/\beta$, which guarantees that its Jacobian $\mathbf{J}_{\boldsymbol{\sigma}}$ is symmetric and positive semidefinite. Furthermore, as proved in [28], gradient networks (10) with softmax activation can universally approximate any continuous monotone gradient field.

As a computationally efficient alternative, we propose the p-norm gradient activation shown in Fig. 4(b), defined as

$$\boldsymbol{\sigma}(\mathbf{z}) = \frac{1}{\left[1 + \sum_{n=1}^{N} (\beta z_n)^p\right]^{\frac{p-1}{p}}} \begin{bmatrix} (\beta z_1)^{p-1} \\ \vdots \\ (\beta z_N)^{p-1} \end{bmatrix} \tag{14}$$

where p is a positive even integer and $\beta > 0$ is a learnable parameter. This activation is the gradient of the smooth regularized p-norm $S(\mathbf{z}) = [1 + \sum_{n=1}^{N} (\beta z_n)^p]^{(1/p)}/\beta$. The convexity of $S(\mathbf{z})$ guarantees monotonicity, while the unit term ensures differentiability at the origin and prevents division by zero. When p is chosen as a power of two, transcendental functions are avoided completely: powers and fractional roots reduce to repeated multiplications and square roots, whereas the softmax requires N exponential evaluations per point.[1] In magnetic modeling, the choice $p = 8$ also aligns with classical empirical exponents used to describe iron saturation curves [3], [21], [31].

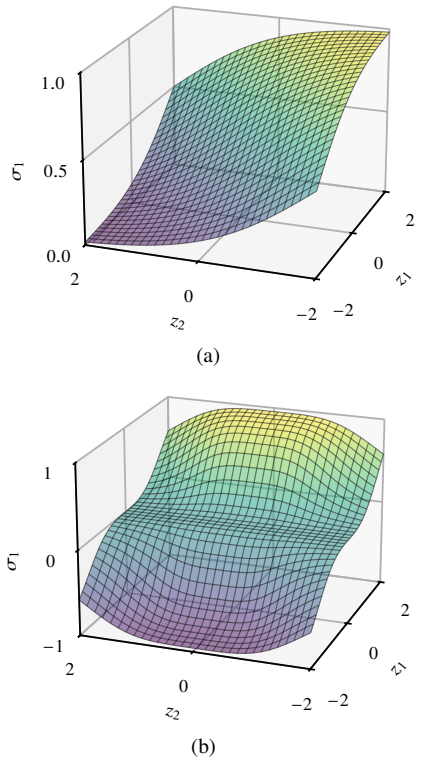


Fig. 4. Vector activation $\sigma_1(z_1, z_2)$ visualized in two-dimensional case: (a) softmax (13) with $\beta = 1$; (b) p-norm gradient (14) with $p = 4$ and $\beta = 1$.

Vector activations are general-purpose: both the softmax (13) and the p-norm gradient (14) apply to current and flux-linkage maps regardless of rotor-angle dependence. While the softmax provides theoretical universality, the proposed p-norm gradient yields comparable accuracy and robustness in practice while significantly reducing the computational burden for real-time execution.

B. Magnetic Model Architectures

This subsection presents the proposed gradient-network architectures in rotor coordinates, using the energy-based current map as the primary example. The same formulation applies analogously to the co-energy-based flux-linkage map.

1) Without Spatial Harmonics: When spatial harmonics are omitted, the field energy depends only on the flux linkage. Without the symmetrized energy (9), the current map can be parametrized directly as $\mathbf{i}_\mathrm{s} = \mathbf{g}(\boldsymbol{\psi}_\mathrm{s})$ using the monotone gradient network (10). Setting $\mathbf{A}_0 = \mathrm{diag}(\mu_\mathrm{d}, \mu_\mathrm{q})$ enforces

[1]For example, for $p = 8$, the fractional power is $S^{7/8} = (\sqrt{\sqrt{\sqrt{S}}})^7$. The flux-linkage maps with the softmax (13) and p-norm gradient (14) activations were compiled to C code and profiled. Profiling indicates that the p-norm gradient executes over three times faster than the softmax across different network sizes.

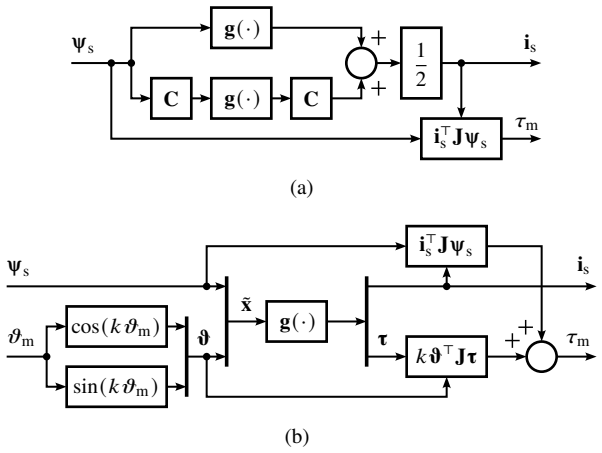


Fig. 5. Proposed magnetic models in rotor coordinates: (a) without spatial harmonics, q-axis symmetry by construction; (b) with spatial harmonics, shown without symmetrization for simplicity. In both cases, the function $\mathbf{g}(\cdot)$ is given by (10).

strong convexity with respect to the flux linkage. The parameters $\mu_\mathrm{d} > 0$ and $\mu_\mathrm{q} > 0$ provide lower bounds for the incremental inverse inductances and are trainable, subject to a small positive lower bound. Consequently, the Jacobian of the current map is positive definite, making the map strongly monotone [30] and robustly invertible to obtain the flux linkage from the current.

With the symmetrized energy function (9), the current map becomes

$$\mathbf{i}_\mathrm{s}(\boldsymbol{\psi}_\mathrm{s}) = \frac{1}{2}\left[\mathbf{g}(\boldsymbol{\psi}_\mathrm{s}) + \mathbf{C}\,\mathbf{g}(\mathbf{C}\boldsymbol{\psi}_\mathrm{s})\right] \tag{15}$$

which enforces q-axis symmetry (8) by construction while preserving the positive definiteness of the Jacobian and the resulting invertibility. Fig. 5(a) shows the corresponding block diagram.

2) With Spatial Harmonics: Fig. 5(b) shows the block diagram for the magnetic model with spatial harmonics, shown without the symmetrization step for simplicity. The combined input to the gradient network is $\tilde{\mathbf{x}} = [\boldsymbol{\psi}_\mathrm{s}^\top, \boldsymbol{\vartheta}^\top]^\top$, and its four outputs correspond to the gradients with respect to $\boldsymbol{\psi}_\mathrm{s}$ and $\boldsymbol{\vartheta}$. Strong monotonicity with respect to the flux linkage is enforced by setting $\mathbf{A}_0 = \mathrm{diag}(\mu_\mathrm{d}, \mu_\mathrm{q}, 0, 0)$, leaving the periodic angle features unconstrained.

Analogously to (15), the symmetrized energy $\overline{W}(\boldsymbol{\psi}_\mathrm{s}, \boldsymbol{\vartheta})$ in (9) is implemented by symmetrizing the network outputs,

$$\begin{bmatrix}\mathbf{i}_\mathrm{s}\\ \boldsymbol{\tau}\end{bmatrix} = \frac{1}{2}\left[\mathbf{g}(\tilde{\mathbf{x}}) + \tilde{\mathbf{C}}\,\mathbf{g}(\tilde{\mathbf{C}}\tilde{\mathbf{x}})\right] \tag{16}$$

where $\tilde{\mathbf{C}} = \mathrm{diag}(\mathbf{C}, \mathbf{C})$ and $\boldsymbol{\tau} = [\partial\overline{W}(\boldsymbol{\psi}_\mathrm{s}, \boldsymbol{\vartheta})/\partial\boldsymbol{\vartheta}]^\top$. This guarantees q-axis symmetry (8) by construction, while preserving strict monotonicity in $\boldsymbol{\psi}_\mathrm{s}$ for each ϑ_m and periodicity in the rotor angle.

IV. Results

Measured and FEM datasets from a four-pole 5.6-kW PM synchronous reluctance machine (ABB Baldor ECS101 M0 H7 EF4) were used to validate the proposed magnetic models. Table I lists the rated values of the machine, and Fig. 6 shows the test bench used for measurements. Due to the rotor flux barriers, a small effective air gap in the q-axis direction, and the presence of PMs, the machine has highly nonlinear magnetic characteristics. While omitted here due to space constraints, flux-linkage maps were also measured for an automotive 10-pole 28-kW PM synchronous reluctance machine (Brusa HSM1.10.18.04) and fitted with similar accuracy.[2]

TABLE I
Rated Values of the 5.6-kW PM Synchronous Reluctance Machine

Voltage (line-to-neutral, peak value)	$\sqrt{2/3}\cdot 460$ V	1.00 p.u.
Current (peak value)	$\sqrt{2}\cdot 8.8$ A	1.00 p.u.
Frequency	60 Hz	1.00 p.u.
Speed	1 800 r/min	1.00 p.u.
Power	5.6 kW	0.80 p.u.
Torque	29.7 Nm	0.80 p.u.

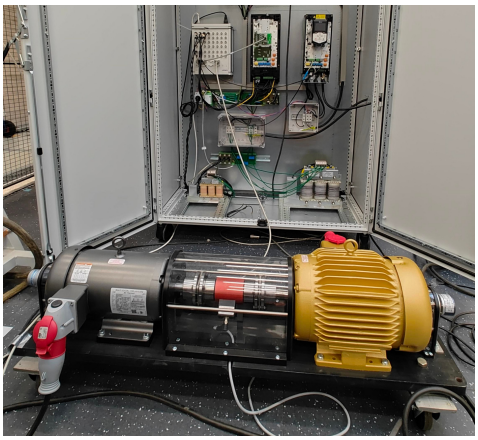


Fig. 6. Test bench including a 5.6-kW PM synchronous reluctance machine (left), load machine (right), and their inverter cabinet (background).

Training was implemented using the *PyTorch* library and *AdamW* optimizer until the training loss plateaued, with learning rate 10^{-3}, batch size 128, and random seed 42. The number of epochs was 20 000 for the measured dataset and 1 000 for the FEM dataset.

A. Measured Dataset Without Spatial Harmonics

1) Dataset: The flux linkages of the example machine were measured on an equidistant current grid using the constant-speed test [20]. By exploiting q-axis symmetry in the measurement procedure, the dataset contains 21×13 unique measurement points. Both d- and q-axis flux linkages were measured at each point. Figs. 7 and 8 show this dataset in flux-linkage coordinates and current coordinates, respectively. For illustration, both positive and negative q-axis values are shown, yielding a full grid of 21×27 points in each map (567 displayed values per map).

2) Model Configuration: Both energy-based current maps $\mathbf{i}_\mathrm{s}(\boldsymbol{\psi}_\mathrm{s})$ and co-energy-based flux-linkage maps $\boldsymbol{\psi}_\mathrm{s}(\mathbf{i}_\mathrm{s})$ were trained using the proposed approach. The q-axis symmetry

[2]Both 5.6-kW and 28-kW machine datasets, training details, and models are available in the *motulator* open-source project: https://aalto-electric-drives.github.io/.

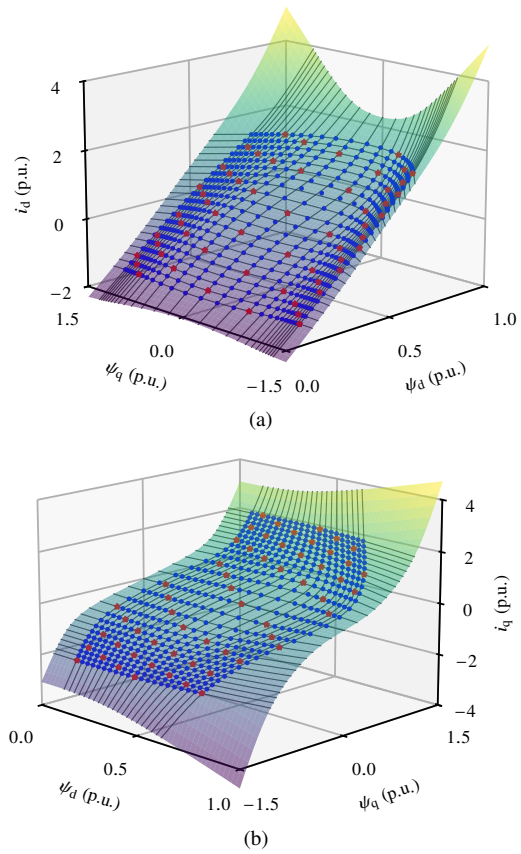


Fig. 7. Current maps: (a) $i_\mathrm{d}(\psi_\mathrm{d}, \psi_\mathrm{q})$; (b) $i_\mathrm{q}(\psi_\mathrm{d}, \psi_\mathrm{q})$. The surfaces show the predicted maps from the squareplus gradient model (11) with $N = 12$ hidden units, trained on a 10% subset. Markers show the measured dataset: red indicates the 10% training subset and blue the remaining points. Gray lines show constant-current contours corresponding to the measured dataset.

TABLE II
COMPARISON OF ACTIVATION FUNCTIONS IN CURRENT MAPS WITHOUT SPATIAL HARMONICS ($N = 12$)

Activation	Training data	e_rms (p.u.)	e_max (p.u.)	e_std (p.u.)
squareplus (11)	10%	0.017	0.070	0.011
	2%	0.076	0.344	0.054
softmax (13)	10%	0.031	0.226	0.021
	2%	0.108	0.407	0.068
p-norm gradient (14)	10%	0.021	0.110	0.012
	2%	0.096	0.389	0.061

is enforced by construction, see Fig. 5(a). Different activation functions (11)–(14) are compared.

The magnetic models without spatial harmonics have two scalar inputs and two scalar outputs. The number of learnable parameters depends on the number N of hidden units. The models have an $N\times2$ weight matrix $\mathbf{A}$, an $N\times1$ bias vector $\mathbf{b}$, a 2×2 diagonal output matrix $\mathbf{A}_0$, a 2×1 output bias $\mathbf{b}_0$, and a single learnable activation parameter β. Consequently, the total number of learnable parameters is $3N+5$. In the following, we use $N = 12$, resulting in 41 learnable parameters. This number of parameters is small compared to the lookup-table approach.

The mean squared error (MSE) loss was used for training

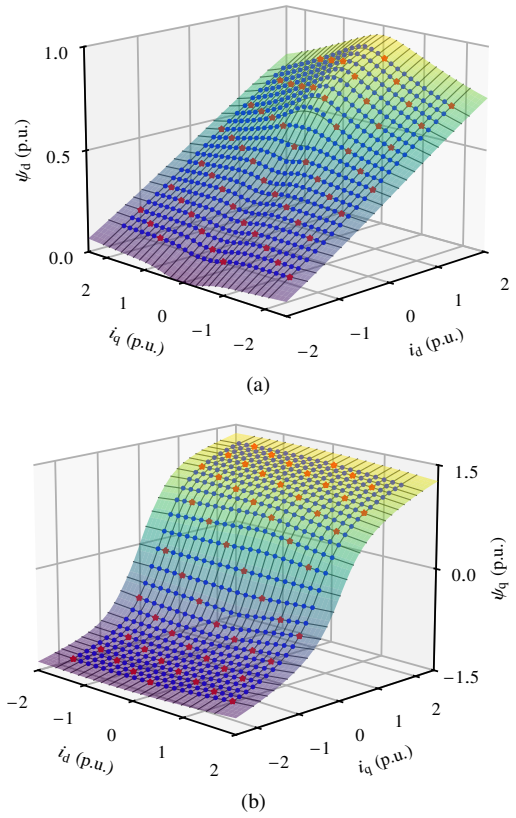


Fig. 8. Flux-linkage maps: (a) $\psi_\mathrm{d}(i_\mathrm{d}, i_\mathrm{q})$; (b) $\psi_\mathrm{q}(i_\mathrm{d}, i_\mathrm{q})$. The surfaces show the predicted maps from the p-norm gradient model (14) with $p = 8$ and $N = 12$ hidden units, trained on a 10% subset. Markers show the same measured dataset as in Fig. 7: red indicates the 10% training subset and blue the remaining points. Gray lines show constant-current contours corresponding to the measured dataset.

TABLE III
COMPARISON OF ACTIVATION FUNCTIONS IN FLUX-LINKAGE MAPS WITHOUT SPATIAL HARMONICS ($N = 12$)

Activation	Training data	e_rms (p.u.)	e_max (p.u.)	e_std (p.u.)
algebraic sigmoid (12)	10%	0.016	0.044	0.010
	2%	0.051	0.165	0.032
softmax (13)	10%	0.007	0.033	0.004
	2%	0.029	0.081	0.019
p-norm gradient (14)	10%	0.004	0.022	0.003
	2%	0.018	0.061	0.012

current maps, defined as

$$\mathcal{L} = \frac{1}{L}\sum_{\ell=1}^{L}\left\|\mathbf{i}_{\mathrm{s}\ell} - \hat{\mathbf{i}}_{\mathrm{s}\ell}\right\|^2 \tag{17}$$

where L is the number of training samples, $\mathbf{i}_{\mathrm{s}\ell}$ is the measured current for sample ℓ, and $\hat{\mathbf{i}}_{\mathrm{s}\ell}(\boldsymbol{\psi}_{\mathrm{s}\ell})$ is the model prediction for the corresponding flux-linkage input $\boldsymbol{\psi}_{\mathrm{s}\ell}$. For training flux-linkage maps, the same procedure was used, replacing the current error with the flux-linkage error.

3) Model Performance: Fig. 7 visualizes the learned current maps with the squareplus activation (11), when every tenth data point of the full measured dataset is used for training and the rest for validation. Fig. 8 shows flux-linkage maps learned by training the co-energy-based model with the p-

TABLE IV
COMPARISON OF ACTIVATION FUNCTIONS WITH SPATIAL HARMONICS ($N = 48$) FOR TORQUE ERROR

Activation	Training data	e_{rms} (p.u.)	$e_{\max}$ (p.u.)	e_{std} (p.u.)
softmax (13)	10%	0.011	0.062	0.007
	2%	0.015	0.091	0.010
p-norm gradient (14)	10%	0.015	0.106	0.010
	2%	0.025	0.140	0.016

norm gradient activation (14) and the same training dataset. To facilitate visual comparison, constant-current contours corresponding to the measurement grid are drawn on the predicted surfaces. Ideally, the measured points would fall exactly at the intersections of these contours. It can be observed that the model very accurately captures the measured data. The other activation functions yield visually similar results when trained with 10% of the dataset.

For quantitative comparison, the root-mean-square (rms) error and the standard deviation over the entire measured dataset are computed, respectively, as

$$e_{\text{rms}} = \sqrt{\frac{1}{L}\sum_{\ell=1}^{L} e_\ell^2} \qquad e_{\text{std}} = \sqrt{\frac{1}{L}\sum_{\ell=1}^{L} (e_\ell - \bar{e})^2} \tag{18}$$

where $e_\ell = \|\mathbf{i}_{\text{s}\ell} - \hat{\mathbf{i}}_{\text{s}\ell}\|$ and $\bar{e} = \frac{1}{L}\sum_\ell e_\ell$. Furthermore, the maximum error $e_{\max} = \max_\ell e_\ell$ is considered. The above metrics are defined for current maps, but they are similarly defined for flux-linkage maps.

Table II summarizes the results for current maps when different activation functions are used. The table shows the results for two different training dataset sizes, 10% and 2% of the full measured dataset. In the latter case, every 50th data point is used for training, resulting in only 12 training samples per map. The results show that all models achieve excellent accuracy when trained with 10% of the dataset and good results even with 2% of the dataset. These results indicate that the underlying energy function can be well approximated with a sum of ridge functions.

Table III gives the results for flux-linkage maps. The models with vector activations show better accuracy than those with elementwise activations, especially when the training dataset is limited to 2%. These results suggest that the underlying co-energy function may not be well approximated by a sum of ridge functions, thus requiring vector activations for accurate modeling with limited data.

To evaluate physical consistency, the proposed gradient-network current map (squareplus activation, $N = 12$, 41 parameters) was compared with an unconstrained multilayer perceptron (MLP) using the same activation and hidden-layer size (65 parameters). With 2% training data, the proposed model gives lower validation error than the MLP ($e_{\text{rms}} = 0.095$ p.u. vs. 0.241 p.u.). The mean reciprocity residual $|\Gamma_{\text{dq}} - \Gamma_{\text{qd}}|$ of the Jacobian $\mathbf{\Gamma}_{\text{s}} = \partial \mathbf{i}_{\text{s}}/\partial \boldsymbol{\psi}_{\text{s}}$ stays at single-precision round-off for the gradient network, while the MLP strongly violates reciprocity ($6.8 \cdot 10^{-8}$ p.u. vs. 0.62 p.u.).

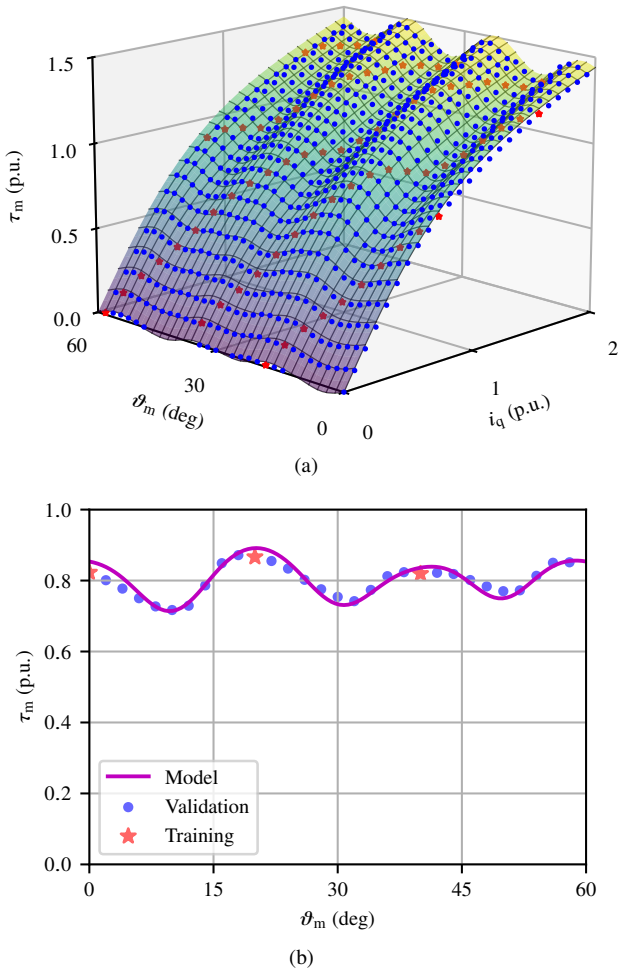


Fig. 9. Electromagnetic torque from the p-norm gradient model with $N = 48$ hidden units: (a) $\tau_{\text{m}}(i_{\text{q}}, \vartheta_{\text{m}})$ at constant i_{d} corresponding to the rated MTPA current; (b) $\tau_{\text{m}}(\vartheta_{\text{m}})$ at the constant rated MTPA stator current. The curve and markers in (b) are a slice from (a). Red and blue markers show the full FEM dataset. Red markers indicate the 10% training subset.

B. FEM Dataset With Spatial Harmonics

1) Dataset: To demonstrate the capability of the proposed model in Fig. 5(b) in capturing spatial harmonics, the 5.6-kW machine was analyzed using FEM under magnetostatic conditions. For this machine, both the flux linkage and torque are periodic in the electrical angle ϑ_{m} with a 60° period. The FEM dataset was generated on an equidistant current grid (61× 61), ranging from −2.41 to 2.41 p.u., and at 30 equidistant rotor angles from 0° to 60°. The total dataset size is $61 \times 61 \times 30 = 111\,630$ samples. At each operating point, the d- and q-axis flux linkages and the electromagnetic torque were computed.

2) Model Configuration: Both the energy-based and co-energy-based models were trained using the proposed approach, with q-axis symmetry enforced by construction via (16). For brevity, the results are shown only for the co-energy-based model, but the energy-based model shows similar performance. The vector activations softmax (13) and the p-norm gradient (14) are compared. The training dataset size is either 10% or 2% of the full FEM dataset, corresponding to 11 163 and 2 232 samples, respectively.

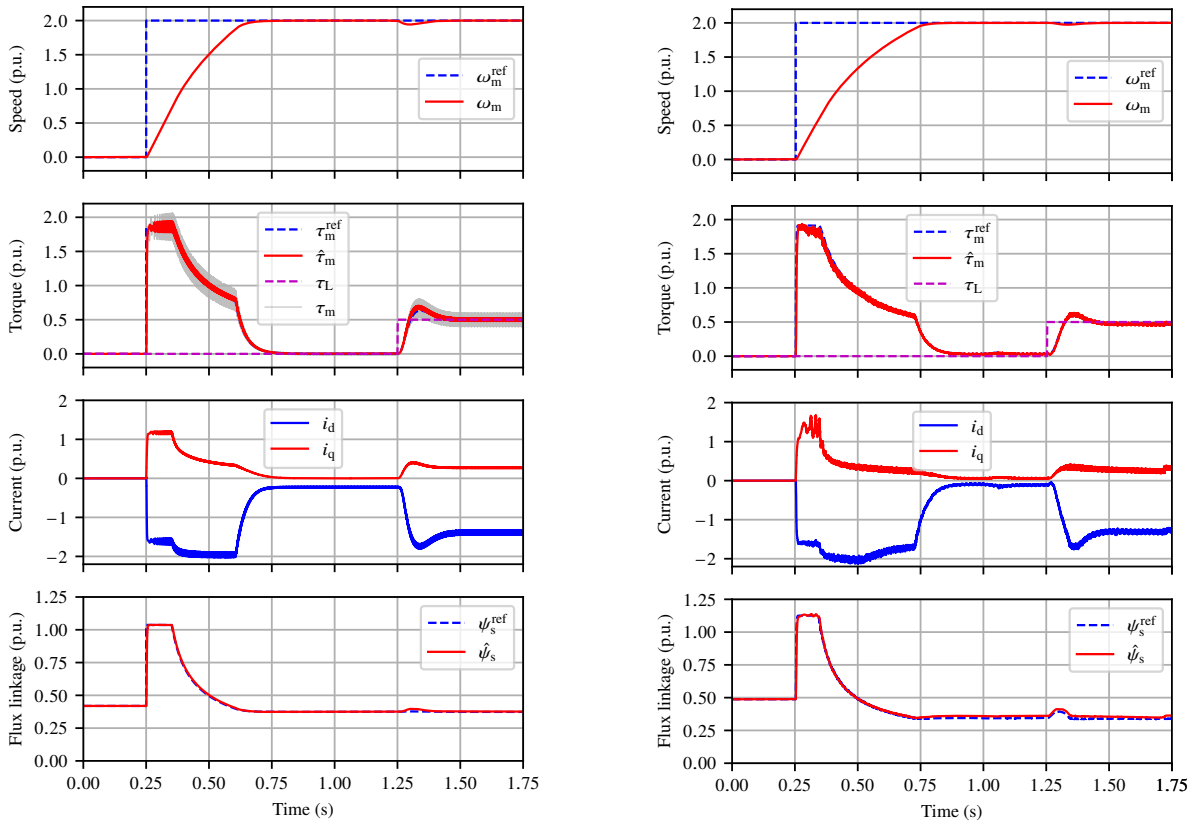


Fig. 10. Closed-loop control responses: (a) simulation with the proposed magnetic model including spatial harmonics in the plant; (b) corresponding measurement. The same controller is used in both cases, parameterized entirely by the proposed model without spatial harmonics. Speed reference step from 0 to 2 p.u. at $t = 0.25$ s and load-torque step at $t = 1.25$ s. From top to bottom, the responses are speed, torque, current, and flux linkage.

The magnetic models with spatial harmonics have four scalar inputs and four scalar outputs. In the following examples, $N = 48$ hidden units are used, resulting in $5N + 7 = 247$ learnable parameters. The MSE loss is used for training, but it includes both the flux-linkage and torque errors. In the case of the co-energy-based models, the loss function is defined as

$$\mathcal{L} = \frac{1}{L}\sum_{\ell=1}^{L}\left[\frac{\left\|\boldsymbol{\Psi}_{\mathrm{s}\ell} - \hat{\boldsymbol{\Psi}}_{\mathrm{s}\ell}\right\|^2}{\psi_{\max}^2} + \frac{\left(\tau_{\mathrm{m}\ell} - \hat{\tau}_{\mathrm{m}\ell}\right)^2}{\tau_{\max}^2}\right] \tag{19}$$

where $\psi_{\max}$ and $\tau_{\max}$ are the maximum flux linkage and torque in the training dataset, respectively. When training energy-based models, the loss function is defined similarly, replacing the flux linkage error term with the current error term.

3) Model Performance: Fig. 9(a) visualizes the proposed co-energy-based model with the p-norm gradient activation (14), trained with 10% of the FEM dataset. The model captures both the rotor-angle dependence and the q-axis current dependence of the torque. Fig. 9(b) shows a slice at $i_{\mathrm{d}} = -0.72$ p.u. and $i_{\mathrm{q}} = 0.72$ p.u., which approximately corresponds to the rated maximum-torque-per-ampere (MTPA) current. The torque ripple caused by spatial harmonics is well captured.

Table IV summarizes the torque prediction errors of the co-energy-based model over the entire FEM dataset. The model achieves good accuracy even when the training dataset is limited to 2% of the full FEM dataset. The p-norm gradient activation (14) gives results comparable to the softmax activation (13) when trained with 10% of the dataset, but yields slightly higher errors when trained with 2% of the dataset. The flux-linkage errors (not shown for brevity) follow the same trend: the softmax is slightly better than the p-norm gradient, and both activations degrade only moderately when reducing the training set from 10% to 2%.

C. Application Examples

To demonstrate the practical applicability of the proposed approach, the learned magnetic models are evaluated in high-fidelity simulations and experimentally on a real-time closed-loop drive system. Fig. 6 shows the experimental setup, where the same 5.6-kW PM synchronous reluctance machine is controlled by a dSPACE MicroLabBox.

The controller is a flux-vector control scheme [33], [34] parameterized by the proposed co-energy-based model (6) without spatial harmonics [p-norm gradient activation (14) with $p = 8$, $N = 6$]. The control system comprehensively employs the proposed model in its control law, flux observer, and online reference generation. Consequently, the model is evaluated multiple times during each 12-kHz control cycle. Notably, the applied online reference generation method [35] numerically inverts the flux-linkage map to obtain the current for the maximum-torque-per-volt (MTPV) calculation, which

requires the magnetic model to be physically consistent and invertible.

1) High-Fidelity Simulation: In the simulation, the machine is represented by the proposed energy-based magnetic model (7) with spatial harmonics (softmax activation, $N = 48$), trained on the FEM dataset as described in Section IV-B. This model variant is convenient since the flux linkages are the state variables of the voltage equations (4). Fig. 10(a) shows the corresponding response. In the shown sequence, the drive operates in the MTPA, field-weakening, current-limit, and MTPV regions. Spatial-harmonic effects (the sixth electrical harmonic and its multiples) are most visible in the actual torque, but also appear in other signals.

2) Real-Time Control Experiment: Fig. 10(b) shows the corresponding measurement using the same control method as in the simulation. In addition to the expected sixth electrical harmonic and its multiples, the measured q-axis current contains first- and fourth-order mechanical harmonics. These low-order components are likely caused by practical asymmetries (e.g., eccentricity, ovality, or winding imbalance) not present in the symmetric FEM model. If necessary, such phenomena could be incorporated into the proposed model by introducing corresponding physical variables, such as lateral rotor displacement, as additional inputs.

Additionally, transients appear in the flux reference during field weakening due to actual DC-bus voltage variations, whereas the simulation assumed a constant DC-bus voltage. Otherwise, the simulation and measurement agree well (although the FEM dataset naturally only approximates the real machine because of uncertainties in geometry and material properties). Most importantly, the experiment confirms that the proposed gradient-network model executes seamlessly as an integral part of the real-time control system.

V. Conclusions

A physics-constrained magnetic modeling framework was presented for synchronous machines. In this framework, the magnetic characteristics and electromagnetic torque are modeled directly as gradients of a field energy or co-energy function. This structure ensures energy conservation and reciprocity by construction. Spatial harmonics are captured by encoding the rotor angle through periodic Fourier features. Monotone gradient networks ensure a unique and invertible relationship between flux linkages and currents. Symmetrizing the energy functions enforces q-axis symmetry by construction. An efficient p-norm gradient activation reduces the computational cost of real-time execution. The framework was validated using measured and FEM datasets, demonstrating high accuracy and data efficiency. Furthermore, closed-loop experiments on an embedded platform confirmed its real-time applicability.

Appendix A
Transformation to Rotor Coordinates

The same field energy can be expressed in different coordinates, $W(\boldsymbol{\psi}_\mathrm{s}, \vartheta_\mathrm{m}) = W^\mathrm{s}(\boldsymbol{\psi}_\mathrm{s}^\mathrm{s}, \vartheta_\mathrm{m})$. Using the chain rule with $\boldsymbol{\psi}_\mathrm{s}^\mathrm{s} = \mathrm{e}^{\vartheta_\mathrm{m}\mathbf{J}}\boldsymbol{\psi}_\mathrm{s}$ gives

$$\frac{\partial W(\boldsymbol{\psi}_\mathrm{s}, \vartheta_\mathrm{m})}{\partial \boldsymbol{\psi}_\mathrm{s}} = \frac{\partial W^\mathrm{s}(\boldsymbol{\psi}_\mathrm{s}^\mathrm{s}, \vartheta_\mathrm{m})}{\partial \boldsymbol{\psi}_\mathrm{s}^\mathrm{s}} \frac{\partial \boldsymbol{\psi}_\mathrm{s}^\mathrm{s}}{\partial \boldsymbol{\psi}_\mathrm{s}} = (\mathbf{i}_\mathrm{s}^\mathrm{s})^\top \mathrm{e}^{\vartheta_\mathrm{m}\mathbf{J}} = \mathbf{i}_\mathrm{s}^\top \tag{20a}$$

$$\begin{aligned}\frac{\partial W(\boldsymbol{\psi}_\mathrm{s}, \vartheta_\mathrm{m})}{\partial \vartheta_\mathrm{m}} &= \frac{\partial W^\mathrm{s}(\boldsymbol{\psi}_\mathrm{s}^\mathrm{s}, \vartheta_\mathrm{m})}{\partial \boldsymbol{\psi}_\mathrm{s}^\mathrm{s}} \frac{\partial \boldsymbol{\psi}_\mathrm{s}^\mathrm{s}}{\partial \vartheta_\mathrm{m}} + \frac{\partial W^\mathrm{s}(\boldsymbol{\psi}_\mathrm{s}^\mathrm{s}, \vartheta_\mathrm{m})}{\partial \vartheta_\mathrm{m}} \\ &= \mathbf{i}_\mathrm{s}^\top \mathbf{J} \boldsymbol{\psi}_\mathrm{s} + \frac{\partial W^\mathrm{s}(\boldsymbol{\psi}_\mathrm{s}^\mathrm{s}, \vartheta_\mathrm{m})}{\partial \vartheta_\mathrm{m}}.\end{aligned} \tag{20b}$$

Using $\tau_\mathrm{m} = -\partial W^\mathrm{s}/\partial \vartheta_\mathrm{m}$ yields the expressions in (5).

Appendix B
Jacobian of the Gradient Network Model

The vector field (10) is conservative (i.e., a gradient field) if its Jacobian $\mathbf{J_g}$ is symmetric [28]. Applying the chain rule, the Jacobian of the gradient network (10) becomes

$$\mathbf{J_g}(\mathbf{x}) = \frac{\partial \mathbf{g}(\mathbf{x})}{\partial \mathbf{x}} = \mathbf{A}_0 + \mathbf{A}^\top \frac{\partial \boldsymbol{\sigma}(\mathbf{z})}{\partial \mathbf{z}} \mathbf{A}. \tag{21}$$

The first term $\mathbf{A}_0$ is chosen to be symmetric. The second term is symmetric if $\mathbf{J_\sigma} = \partial \boldsymbol{\sigma}/\partial \mathbf{z}$ is symmetric, which is the case when the activation $\boldsymbol{\sigma}$ is the gradient of a scalar function. Note that $\mathbf{J_\sigma}$ is diagonal and thus symmetric when the activation is elementwise. The network (10) is monotone if $\mathbf{J_g}$ is positive semidefinite. This is guaranteed if $\mathbf{A}_0$ and $\mathbf{J_\sigma}$ are positive semidefinite.

References

[1] H. H. Woodson and J. R. Melcher, *Electromechanical Dynamics*. John Wiley & Sons, 1968. [Online]. Available: https://ocw.mit.edu/ans7870/resources/woodson/textbook/emd_part1.pdf

[2] A. E. Fitzgerald, C. Kingsley, Jr., and S. D. Umans, *Electric Machinery*, 6th ed. Boston, MA: McGraw-Hill, 2003.

[3] Z. Qu, T. Tuovinen, and M. Hinkkanen, "Inclusion of magnetic saturation in dynamic models of synchronous reluctance motors," in *Proc. Int. Conf. Electrical Machines*, Marseille, France, Sep. 2012, pp. 994–1000.

[4] M. Gärtner, F. Schuller, N. Parspour, and P. Seibold, "Analytical modeling and simulation of highly utilized electrical machines considering nonlinear effects," in *Proc. Int. Conf. Elect. Machines*, Marseille, France, Sep. 2012, pp. 2786–2791.

[5] S.-W. Su, C. M. Hackl, and R. Kennel, "Analytical prototype functions for flux linkage approximation in synchronous machines," *IEEE Open J. Ind. Electron. Soc.*, vol. 3, pp. 265–282, 2022.

[6] A. Varatharajan, P. Pescetto, S. Ferrari, and G. Pellegrino, "Flux maps spatial harmonic modeling and measurement in synchronous reluctance motors," in *Proc. IEEE ECCE*, Nashville, TN, Oct./Nov. 2023, pp. 4868–4873.

[7] F. Lelli, M. Hinkkanen, and F. G. Capponi, "A saturation model based on a simplified equivalent magnetic circuit for permanent magnet machines," in *Proc. Int. Conf. Elect. Machines*, Torino, Italy, Sep. 2024.

[8] A. Vagati, M. Pastorelli, G. Franceschini, and V. Drogoreanu, "Flux-observer-based high-performance control of synchronous reluctance motors by including cross saturation," *IEEE Trans. Ind. Appl.*, vol. 35, no. 3, pp. 597–605, May/Jun. 1999.

[9] X. Chen, J. Wang, B. Sen, P. Lazari, and T. Sun, "A high-fidelity and computationally efficient model for interior permanent-magnet machines considering the magnetic saturation, spatial harmonics, and iron loss effect," *IEEE Trans. Ind. Electron.*, vol. 62, no. 7, pp. 4044–4055, Jul. 2015.

[10] S. Li, D. Han, and B. Sarlioglu, "Modeling of interior permanent magnet machine considering saturation, cross coupling, spatial harmonics, and temperature effects," *IEEE Trans. Transport. Electrific.*, vol. 3, no. 3, pp. 682–693, Sep. 2017.

[11] J. Lee, Y.-C. Kwon, and S.-K. Sul, "Identification of IPMSM flux-linkage map for high-accuracy simulation of IPMSM drives," *IEEE Trans. Power Electron.*, vol. 36, no. 12, pp. 14 257–14 266, Dec. 2021.

[12] S. Ferrari, G. Dilevrano, P. Ragazzo, and G. Pellegrino, "The dq-theta flux map model of synchronous machines," in *Proc. IEEE ECCE*, Vancouver, Canada, Oct. 2021, pp. 3716–3723.

[13] A. Bojoi, P. Pescetto, S. Ferrari, and G. Pellegrino, "Experimental identification of the dqθ flux maps of synchronous machines," in *Proc. Int. Conf. Elect. Machines*, Turin, Italy, Sep. 2024.

[14] M. Jastrzębski and J. Kabziński, "Approximation of permanent magnet motor flux distribution by partially informed neural networks," *Energies*, vol. 14, no. 18, 2021.

[15] D. Pasqualotto, S. Rigon, and M. Zigliotto, "Sensorless speed control of synchronous reluctance motor drives based on extended Kalman filter and neural magnetic model," *IEEE Trans. Ind. Electron.*, vol. 70, no. 2, pp. 1321–1330, Feb. 2023.

[16] Z. Liu and S. Li, "Flux linkage identification of IPM motor through neural-network considering speed impact," in *Proc. IEEE Power and Energy Society General Meeting (PESGM)*, Seattle, WA, Jul. 2024.

[17] J.-H. Lee, Y.-J. Lee, M.-S. Lee, D.-S. Jin, and Y.-D. Yoon, "Estimation of flux saturation model for SynRMs using artificial neural network," *IEEE Trans. Ind. Appl.*, vol. 61, no. 2, pp. 3143–3151, Mar./Apr. 2025.

[18] Z. Shao, Y. Bao, Z. Zhong, Z. Ma, and Y. Wang, "Neural network-based distributed parameter modeling of permanent magnet synchronous machines considering spatial harmonics and iron losses," *J. Power Electron.*, Feb. 2026.

[19] G. Y. Sizov, D. M. Ionel, and N. A. O. Demerdash, "Modeling and parametric design of permanent-magnet AC machines using computationally efficient finite-element analysis," *IEEE Trans. Ind. Electron.*, vol. 59, no. 6, pp. 2403–2413, Jun. 2012.

[20] E. Armando, R. Bojoi, P. Guglielmi, G. Pellegrino, and M. Pastorelli, "Experimental identification of the magnetic model of synchronous machines," *IEEE Trans. Ind. Appl.*, vol. 49, no. 5, pp. 2116–2125, Sep. 2013.

[21] M. Hinkkanen, P. Pescetto, E. Mölsä, S. E. Saarakkala, G. Pellegrino, and R. Bojoi, "Sensorless self-commissioning of synchronous reluctance motors at standstill without rotor locking," *IEEE Trans. Ind. Appl.*, vol. 53, no. 3, pp. 2120–2129, May/Jun. 2017.

[22] P. Sauer, "Constraints on saturation modeling in AC machines," *IEEE Trans. Energy Convers.*, vol. 7, no. 1, pp. 161–167, Mar. 1992.

[23] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

[24] M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho, "Lagrangian neural networks," in *ICLR Workshop on Integration of Deep Neural Models and Differential Equations*, 2020. [Online]. Available: https://arxiv.org/abs/2003.04630

[25] S. Greydanus, M. Dzamba, and J. Yosinski, "Hamiltonian neural networks," in *Adv. Neur. Inf. Process. Syst.*, vol. 32, Vancouver, Canada, 2019. [Online]. Available: https://arxiv.org/abs/1906.01563

[26] S. A. Desai, M. Mattheakis, D. Sondak, P. Protopapas, and S. J. Roberts, "Port-Hamiltonian neural networks for learning explicit time-dependent dynamical systems," *Phys. Rev. E*, vol. 104, p. 034312, Sep. 2021.

[27] A. van der Schaft and D. Jeltsema, *Port-Hamiltonian Systems Theory: An Introductory Overview*. Delft, Netherlands: now Publishers Inc., 2014.

[28] S. Chaudhari, S. Pranav, and J. M. F. Moura, "Gradient networks," *IEEE Trans. Signal Process.*, vol. 73, pp. 324–339, 2025.

[29] M. Tancik, P. Srinivasan, B. Mildenhall, S. Fridovich-Keil, N. Raghavan, U. Singhal, R. Ramamoorthi, J. Barron, and R. Ng, "Fourier features let networks learn high frequency functions in low dimensional domains," in *Adv. Neur. Inf. Process. Syst.*, vol. 33, 2020.

[30] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, UK: Cambridge University Press, 2004.

[31] J. Fischer and U. Moser, "Die Nachbildung von Magnetisierungskurven durch einfache algebraische oder transzendente Funktionen," *Arch. für Electrotechnik*, vol. 42, no. 5, pp. 286–299, 1956.

[32] J. T. Barron, "Squareplus: A softplus-like algebraic rectifier," 2021. [Online]. Available: https://arxiv.org/abs/2112.11687

[33] L. Tiitinen, M. Hinkkanen, and L. Harnefors, "Design framework for sensorless control of synchronous machine drives," *IEEE Trans. Ind. Electron.*, vol. 72, no. 2, pp. 1379–1390, Feb. 2025.

[34] A. Varatharajan, G. Pellegrino, and E. Armando, "Direct flux vector control of synchronous motor drives: Accurate decoupled control with online adaptive maximum torque per ampere and maximum torque per volts evaluation," *IEEE Trans. Ind. Electron.*, vol. 69, no. 2, pp. 1235–1243, Feb. 2022.

[35] M. Sarén, H. Hartikainen, A. Piippo, and M. Hinkkanen, "Decoupled online feedforward generation of optimal references for saturated synchronous machine drives," 2026. [Online]. Available: https://arxiv.org/abs/2607.08528



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